

# Eco-Routing in Urban Networks: A Comparative Study of A\* and Dijkstra on Ecology-Aware and Fastest Objectives

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**Abstract**—Road transport is a major contributor to urban energy consumption and greenhouse gas emissions, yet traditional navigation systems often prioritize travel time or distance, neglecting critical factors such as topography. In cities with hilly terrain, such as Divinópolis - MG, road gradients strongly influence fuel usage, meaning the shortest routes are not always the most efficient in terms of energy use. This work aims to develop and validate a deterministic eco-routing framework that minimizes estimated fuel consumption while maintaining competitive travel times, without relying on massive historical Floating Car Data (FCD). The proposed methodology utilizes OpenStreetMap data and digital elevation models to construct a routable graph enriched with a cost function grounded in vehicle dynamics that penalizes steep gradients. Four algorithms were implemented and compared: Eco-Dijkstra, Eco-A\*, Fastest-Dijkstra and Fastest-A\*, employing an admissible heuristic based on the Haversine distance to ensure optimality. Experimental results demonstrate that the ecological route reduces fuel consumption by approximately 5.44% compared to the shortest path. Notably, while the average travel time increased marginally by 1.79%, the analysis indicates that eco-routes can, in certain cases, maintain or even decrease travel duration. Furthermore, the A\* algorithm outperformed Dijkstra in computational efficiency, reducing execution time by about 6.0%. This study concludes that integrating static topographic data into routing algorithms is a viable and scalable strategy for promoting sustainable mobility in medium-sized and developing cities.

**Index Terms**—eco-routing, A\*, Dijkstra, elevation, fuel consumption, route planning

## I. INTRODUCTION

Road transport is a major contributor to urban energy consumption and greenhouse-gas emissions. Traditional route planning systems typically optimize for travel time or distance, but these objectives do not necessarily minimize fuel consumption or emissions. Energy use in vehicles depends on many factors beyond distance, like vehicle speed profile, road grade and stop-and-go dynamics are among the most influential [1]. In hilly urban environments, such as Divinópolis - MG, Brazil, road grade can strongly affect fuel usage: uphill segments increase fuel consumption significantly, while routes with gentler slopes may be more energy-efficient even if they are somewhat longer.

This work addresses the central problem in eco-routing: computing driving routes that minimize estimated fuel consumption while still keeping travel time reasonable. This work presents a practical and reproducible pipeline that builds a routable graph using OpenStreetMap [2] data, enriching it with elevation values, calculating an edge-level ecological cost that blends estimated fuel consumption and a time penalty, and then performing a search aiming for the shortest path, using adapted graph algorithms (A\* and Dijkstra). The dataset for this study is a graph of Divinópolis - MG, obtained and exported to two CSV files (nodes and edges).

Key technical elements of the approach include: (i) a lightweight fuel model per edge that estimates liters consumed as a function of distance, road grade and speed penalties; (ii) a time-penalty term converted to “liters equivalent” so that fuel economy and travel time are combined into a single scalar cost per edge; (iii) preprocessing that computes and sanitizes edge costs (including protections against invalid or negative weights); (iv) two search strategies: a modified Dijkstra that minimizes the combined ecological cost and an A\* variant with an admissible heuristic that lower-bounds the ecological cost using orthodromic distance and the graph’s maximum speed. The slope for each edge is computed robustly from node elevations using a geometric 3D displacement routine to estimate horizontal distance and grade, which yields a more realistic slope estimate than a pure elevation difference divided by stored edge length.

To study both route quality and algorithmic behavior, four routing variants were implemented: eco-Dijkstra and eco-A\* (both minimizing the ecological cost), and fastest-Dijkstra and fastest-A\* (both minimizing travel time). These variants were compared on two levels: (a) routing outcomes: total estimated fuel consumption, total travel time and total distance; (b) algorithmic performance and behavior — runtime, number of explored nodes and sensitivity to the chosen heuristic. In particular, the analysis focuses on how the A\* heuristic affects search efficiency and route selection compared with Dijkstra under the two different optimization objectives (eco x fastest).

Eco-routing is evaluated against standard baselines

(shortest-distance and fastest routes) using metrics that include total estimated fuel consumption, total travel time, and total distance. The eco-routing approach is designed to prefer routes with lower uphill exposure. As a result, it typically reduces the estimated fuel consumption relative to shortest-distance or fastest routes, while introducing only modest increases (or often even reductions) in travel time. Additionally, the sensitivity of the planner to the model parameters (base fuel consumption, slope sensitivity and time-weight) is analyzed, along with a discussion of cases in which eco-routes diverge substantially from conventional routes.

## II. RELATED WORKS

Numerous studies have investigated methodologies for determining eco-routes in urban networks via graph-based approaches. This section highlights key works that align with this study.

### A. Eco-routing problem considering fuel consumption and probabilistic travel time budget

The eco-routing problem has been extensively investigated as a key strategy for reducing greenhouse gas emissions in urban transportation. A recent study by Zeng et al. [3] discusses how traffic condition uncertainty impacts both fuel consumption and travel time, proposing a novel model that integrates these stochastic variables. This aligns with the finding that traditional deterministic algorithms often fail to provide reliable routes in real-world scenarios, where speed variations can significantly compromise both energy efficiency and punctuality.

In their work, Zeng et al. [3] propose an eco-routing framework that minimizes expected fuel consumption subject to a probabilistic travel time budget. Their approach utilizes derived stochastic formulas to estimate fuel consumption distributions based on speed variance, employing an exact algorithm to solve the reliable path problem. This methodology supports the hypothesis that explicitly accounting for uncertainty enhances the potential driver acceptance of eco-routes by ensuring a defined probability of on-time arrival, rather than solely focusing on theoretical fuel savings.

Additionally, Zeng et al. [3] apply their model to a case study using real-world network data, demonstrating the practical applicability of the proposed algorithms. Their study is instrumental in illustrating how incorporating reliability constraints can overcome the limitations of static models, offering a rigorous insight into the trade-offs between fuel economy and temporal efficiency in congested urban networks.

### B. Eco-Friendly Route Planning Algorithms: Taxonomies, Literature Review and Future Directions

A comprehensive survey by Fahmin et al. [4] provides a systematic review of eco-friendly route planning algorithms, organizing the vast body of existing literature into detailed taxonomies. Their work categorizes methods based on critical factors such as vehicle types (internal combustion vs. electric),

objective functions (e.g., fuel vs. energy minimization), and the nature of the road network (static vs. time-dependent).

In their analysis, Fahmin et al. [4] critically evaluate the trade-offs between computational complexity and the accuracy of fuel consumption models used in current routing algorithms. They highlight that while heuristic approaches offer faster solutions for large-scale networks, they often lack the precision required for reliable emissions estimation. Furthermore, the study identifies significant gaps in current research, particularly the lack of standardized benchmarks and the limited integration of real-time traffic uncertainties. Their discussion on future directions provides a roadmap for developing next-generation algorithms that can better handle the dynamic nature of modern urban traffic and emerging vehicle technologies.

### C. Eco-Routing based on a Data Driven Fuel Consumption Model

The development of robust ecological routing strategies is fundamentally dependent on the accuracy of fuel consumption estimation. In a study driven by data, Huang et al. [5] propose a nonparametric fuel consumption model utilizing Gaussian Mixture Regression (GMR) to predict energy costs at the link level. Their research leverages a massive dataset from the Safety Pilot Model Deployment (SPMD) in Ann Arbor, comprising over 3,000 frequent pairs of origin and destination collected from 2,000 vehicles, and utilizes simulations based on physics (Autonomie) to train the model, ensuring it captures the complexities of driving behavior in the real world and road grades.

Addressing the critical balance between energy efficiency and travel duration, Huang et al. [5] introduce an ecological routing algorithm constrained by travel time. This approach employs a soft constraint mechanism to minimize fuel consumption while ensuring the travel time remains within an acceptable threshold relative to the fastest route. The results demonstrate that this constrained strategy achieves significant fuel savings of approximately 5.16% with a negligible increase in travel time (0.91%), offering a more practical solution for driver adoption compared to unconstrained methods that often result in excessive delays.

## III. METHODOLOGY

This study proposes a deterministic approach to the eco-routing problem, prioritizing accessibility and algorithmic efficiency over the reliance on stochastic models driven by massive historical datasets. The methodology is divided into three phases: (i) Collection of data and graph construction using geographic data with open access; (ii) Formulation of a cost function based on physics incorporating topography; and (iii) Implementation of the A\* algorithm with a specific heuristic design to ensure optimality.

### A. Data Collection and Network Modeling

Unlike recent studies that rely on FCD or on extensive historical traffic logs, which are often unavailable in medium

sized cities or in developing regions, this work uses static geometric and topographical data. This choice makes ecological routing solutions more accessible in areas with scarce data. This choice democratizes the application of ecological routing solutions to areas characterized by data scarcity. The road network of Divinópolis - MG, Brazil, was extracted using the *OSMnx* Python library [6], which interfaces with OpenStreetMap. The raw data was processed to construct a directed graph  $G = (V, E)$ , where  $V$  represents intersections (nodes) and  $E$  represents road segments (edges). To incorporate the critical factor of road gradient, which significantly impacts fuel consumption, the graph was enriched using Digital Elevation Models (DEM). Elevation data for each node was fetched via the Open Elevation API [7] using a batch processing approach with caching mechanisms to ensure performance. The slope of each edge  $(u, v)$  was calculated based on the elevation difference  $\Delta h$  and the horizontal distance  $d$ . In the implementation, the horizontal distance  $d_{(u,v)}$  is estimated by a local planar displacement function (`horizontal_displacement_m`) that computes planar displacement components and the resulting horizontal distance in meters. This approximation is suitable for urban segments and consistent with the method used to compute slope in the code, resulting in the following formula:

$$Slope_{(u,v)} = \frac{Elevation_v - Elevation_u}{d_{(u,v)}} \quad (1)$$

Using DEM data, energy costs are analyzed at the road-segment level without relying on instrumented vehicles, thereby addressing the data scarcity that often hinders urban planning in emerging economies.

#### B. Formulation of Cost Function Based on Physics

To determine the ecological cost of a route, the proposed approach moves beyond simple distance minimization. A weighted cost function was calculated in advance for every edge in the graph, considering three main factors: distance, road grade and speed efficiency. The fuel consumption estimation model is deterministic and grounded in vehicle dynamics principles described in [8]. The algorithm penalizes positive gradients (segments with upward inclination) while ignoring negative gradients (downward), reflecting the increased engine load required to overcome gravity. Thus, a simplified cost function defined ( $C_{eco}$ ) for an edge is proposed as:

$$C_{eco} = F_{base} \times (1 + \alpha \cdot \max(0, Slope)) \times V_{factor} + W_{time} \quad (2)$$

Where:

- $F_{base}$  represents the baseline fuel consumption for the segment length on flat terrain.
- $\alpha$  is the slope coefficient, penalizing uphill movement.
- $V_{factor}$  is a penalty factor for speeds deviating from the optimal engine efficiency range.
- $W_{time}$  converts travel time into units equivalent to fuel to balance efficiency and punctuality.

This model ensures that the edge weights in the graph reflect the physical reality of driving, where climbing a steep street

is significantly more demanding on energy resources than traveling the same distance on a flat road.

#### C. Algorithmic Strategy: A\* and Topographical Relaxation

A critical challenge in ecological routing using the A\* algorithm is designing an *admissible heuristic* ( $h(n)$ ). Standard distance heuristics are often insufficient for energy minimization because the shortest path is not necessarily the most efficient in terms of fuel (e.g., a short path over a steep hill versus a longer flat path). However, heuristics that attempt to estimate terrain complexity often overestimate the cost, violating the admissibility condition ( $h(n) \leq h^*(n)$ ) and resulting in suboptimal routes. To solve this, a strategy of Topographical Relaxation is proposed. The routing system operates on two distinct levels:

- 1) **The Physical Reality** ( $g(n)$ ): The accumulated cost from the start node utilizes the full physical model described in Equation 2, heavily penalizing the actual topography.
- 2) **The Heuristic World** ( $h(n)$ ): The estimated cost to the destination assumes an "idealized flat world." The heuristic function calculates the lower bound of fuel consumption assuming a straight line and zero slope [9].

$$h_{eco}(n, target) = F_{base,rate} \times Distance_{haversine}(n, target) \quad (3)$$

By defining the heuristic this way, admissibility is mathematically guaranteed. Since the energy required to travel a distance  $d$  on a flat surface is always less than or equal to the energy required to travel  $d$  with an incline, our heuristic never overestimates the actual cost. This approach provides an elegant balance: it maintains the precision of a physical model for the actual path traversal while retaining the mathematical safety and search efficiency of the A\* algorithm. Consequently, the algorithm guarantees finding the global optimum for fuel consumption typically faster than the Dijkstra algorithm, without requiring complex stochastic formulations.

#### D. Implementation and Visualization

The system was implemented in Python. The A\* and Dijkstra algorithms were developed to run comparatively on the processed graph. To validate the results, a rendering module was built using *Folium*, allowing for the visualization of the "Ecological Route" versus the "Shortest Path" on interactive maps. The system highlights the balance between distance, time and fuel, providing a clear analytical comparison of the algorithmic performance. Finally, a quantitative comparison section was integrated to evaluate the computational efficiency of A\* versus Dijkstra, recording the execution runtime for each path calculation to demonstrate the performance gains of the proposed heuristic.

## IV. RESULTS

In this section, the outcomes derived from a selected test route are presented. To ensure reliability, the program was executed 10 times, and the results for each metric (execution

time, fuel consumption, etc.) were averaged. Two main experiments were conducted: (i) a comparative analysis of route quality between the Ecological (Eco) and Shortest Distance (Shortest) strategies, and (ii) an evaluation of the computational performance between Dijkstra and A\* algorithms.

The underlying graph structure used for these experiments, representing the road network of Divinópolis, is visualized in Fig. 1.

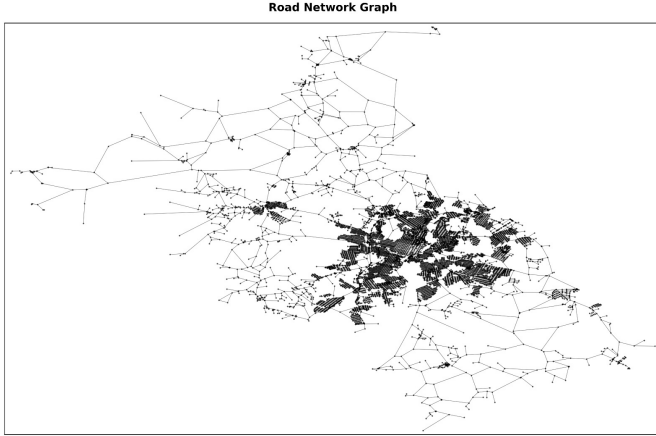


Fig. 1. Visual representation of the directed graph  $G = (V, E)$  constructed for the city of Divinópolis-MG using OpenStreetMap data. The nodes (intersections) and edges (road segments) form the basis for the topographic analysis and route optimization performed in this study.

#### A. Quantitative Analysis of Routes

To validate the fuel consumption model defined in Equation (2), routes were simulated between diverse origin-destination (OD) pairs within the city, specifically targeting points with significant topographic variations.

Table I presents a specific case study for a route calculated between the neighborhood Vila Cruzeiro and the city center, covering approximately 4.5 km.

TABLE I  
COMPARISON BETWEEN SHORTEST PATH AND ECO-ROUTE

| Metric             | Shortest Route | Eco-Route | Difference (%) |
|--------------------|----------------|-----------|----------------|
| Total Distance (m) | 4,457.3        | 4,522.7   | +1.47%         |
| Travel Time (min)  | 5.6            | 5.7       | +1.79%         |
| Fuel (Liters)      | 0.570          | 0.539     | -5.44%         |

As shown in Table I, the Ecological Route resulted in a fuel saving of approximately **5.44%** compared to the Shortest Route. This reduction was achieved with a slight increase in distance (+65.4 meters) and travel time (+0.1 minutes). The trade-off demonstrates that modest deviations from the shortest path can yield significant energy savings when topography is considered.

Figure 2 illustrates the absolute fuel consumption comparison for this test case.

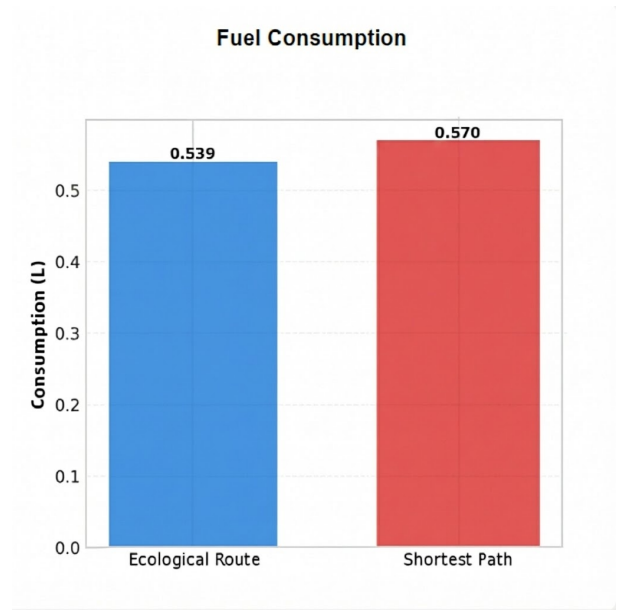


Fig. 2. Quantitative comparison of total estimated fuel consumption (in liters) between the Ecological Route (0.539 L) and the Shortest Path (0.570 L) for the selected test case, highlighting the 5.44% energy saving achieved by the eco-routing algorithm.

#### B. Topographical Influence and Visual Inspection

The efficiency of the Eco-Routing algorithm is a direct result of its design, which penalizes steep positive slopes ( $Slope > 0$ ). Consequently, the algorithm prioritizes flatter segments, avoiding streets with high gradients even if they result in a slightly higher total mileage.

Figure 3 visualizes the resulting paths rendered by the system. The Blue line represents the Shortest Route, while the Red line represents the Ecological Route. The divergence highlights how the ecological algorithm minimizes excessive engine load by avoiding steep ascents.

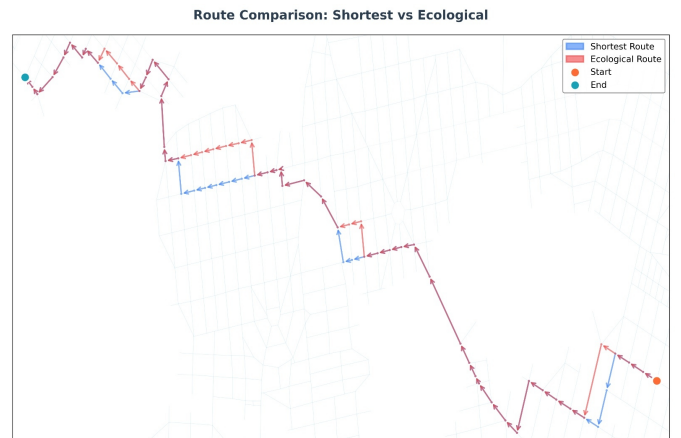


Fig. 3. Geospatial visualization of the route divergence. The Shortest Route (Blue) follows a direct path with steeper gradients, whereas the Ecological Route (Red) selects an alternative path that minimizes positive slope exposure to optimize fuel efficiency, despite a slight increase in distance.

### C. Computational Performance: Dijkstra vs. A\*

The runtime efficiency of both algorithms was evaluated. The A\* algorithm utilized an admissible heuristic based on the Haversine distance and the network's maximum speed to estimate the lower bound of the remaining cost ( $h(n)$ ).

The runtime results summarized in the figure below isolate the pathfinding computational time, excluding data loading and map rendering. The blue bars correspond to Dijkstra's algorithm and the purple bars correspond to the A\* algorithm. Times are reported in milliseconds.

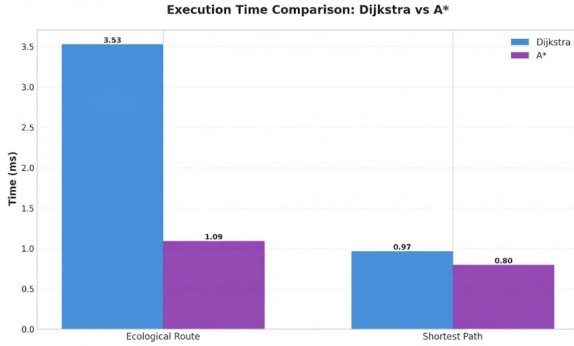


Fig. 4. Execution time comparison (in milliseconds) between Dijkstra and A\* algorithms for calculating Ecological and Shortest routes. The results demonstrate the efficiency gain of the A\* heuristic search, which reduced the average runtime by approximately 6.0% compared to Dijkstra.

The results indicate that the A\* algorithm outperforms Dijkstra in terms of search speed. In our experiments, A\* achieved a runtime reduction of approximately **6.0%** compared to Dijkstra (reducing search time from 6267.47ms to 5893.55ms). This efficiency gain is attributed to the heuristic's ability to direct the search towards the target node, reducing the number of explored edges without compromising the optimality of the solution, as guaranteed by the admissibility of our cost function design.

## V. DISCUSSION

The results of our comparative analysis provide quantitative support for the viability of the topography awareness routing in medium-sized Brazilian cities. The experiments demonstrated that the Eco-Routing strategy effectively reduces estimated fuel consumption (approx. 5.44%) by selecting paths with gentler slopes, often with only a negligible increase in travel distance. Furthermore, the algorithmic evaluation confirms that the A\* algorithm, equipped with our specific heuristic, offers a statistically significant improvement in computational time compared to Dijkstra, making it a scalable solution for real applications.

### A. Limitations

While the results demonstrate the potential of the proposed framework, several limitations inherent to the current model and data availability must be acknowledged:

- **Infrastructure Data Scarcity:** The absence of a detailed registry of pavement types reduces the accuracy of the model. Currently the model assumes a uniform friction coefficient for all edges, which overlooks real variations in rolling resistance across different pavement surfaces. The literature indicates that pavement roughness (commonly quantified by the International Roughness Index — IRI) is positively correlated with increased fuel consumption: rougher pavements cause greater rolling losses and energy dissipation, thereby raising fuel use. Consequently, in urban areas such as Divinópolis, cobblestone (paralelepiped) segments (depending on installation and maintenance) are prone to higher IRI than well-maintained asphalt stretches, and thus can lead to higher fuel consumption, justifying the inclusion of pavement type or measured IRI as a model variable [10].
- **Static Traffic Assumptions:** The proposed cost function does not currently account for real-time traffic conditions or congestion during rush hours. Consequently, the estimated travel times are theoretical calculations based on unimpeded flow and may not reflect the delays incurred during periods of heavy traffic, which would also impact fuel efficiency.
- **Vehicle Dynamics and “Stop-and-Go”:** The physical model assumes a relatively constant speed profile along edges. Energy costs associated with “stop-and-go” dynamics (e.g., acceleration after traffic lights or stop signs), which are known to be major contributors to fuel consumption in urban environments [1], were not incorporated.
- **Visualization Accuracy:** Minor discrepancies were observed in the final map plotting. Due to precision limits in the reverse geocoding and the rendering library, the visual representation of the route overlay may occasionally deviate slightly from the exact physical road geometry, although the underlying graph calculation remains accurate.

### B. Comparison with Existing Literature

Our findings align with previous studies, such as [11] and [12], which argue that minimizing distance is often insufficient for energy optimization in hilly terrains. However, unlike data-driven approaches that require massive historical FCD [11], our method relies solely on static topographic data. This distinction is crucial, as it validates a deterministic, approach based on physics that is applicable in regions where historical traffic data is scarce or unavailable. Our results differ slightly from stochastic models by offering a guaranteed optimal path based on physical constraints rather than probabilistic travel time budgets.

### C. Future Directions

Future studies could extend this work by:

- Incorporating road surface attributes (e.g., tagging OSM data with surface types) to adjust rolling resistance coefficients dynamically.

- Integrating real-time traffic APIs to transition the model from a static graph to a dynamic network, allowing for the avoidance of congestion.
- Implementing a more granular emissions model to account for inertial energy loss during deceleration and acceleration events at intersections.
- Expanding the vehicle model to consider Electric Vehicles (EVs), specifically analyzing the energy recuperation potential (regenerative braking) on the downhill segments identified by the algorithm.
- Scaling the implementation to larger metropolitan areas with dense infrastructure, such as São Paulo. The modular design of the provided source code [13] facilitates adaptation to different urban scales, allowing for the processing of larger graphs and more complex road networks with minimal structural changes.

#### D. Broader Impact

Our approach offers insights into sustainable urban mobility for developing cities with complex topographies. By demonstrating that fuel savings can be achieved without significant time penalties, this methodology provides a foundation for updated GPS navigation systems that prioritize environmental sustainability. Furthermore, it highlights the importance of detailed topographic data in the planning of smart city logistics.

### VI. CONCLUSION

This study presented and validated a deterministic framework for eco-routing in urban networks, using Divinópolis - MG as a case study. By integrating OpenStreetMap data with digital elevation models, a cost function grounded in vehicle dynamics was constructed to penalize routes with steep gradients without relying on massive historical FCD, typically required by data-driven approaches.

The experiments demonstrated the effectiveness of the proposed approach in two main aspects. First, regarding energy efficiency, the Ecological Route achieved an estimated fuel saving of 5.44% compared to the Shortest Path. This reduction was obtained with only marginal penalties in terms of total distance (+1.47%) and travel time (+1.79%), confirming that avoiding aggressive topography is a viable strategy for reducing emissions without significantly compromising urban mobility.

Second, the algorithmic analysis confirmed the superiority of the  $A^*$  algorithm over Dijkstra for this application. The use of an admissible heuristic based on the Haversine distance and maximum road speed allowed for a reduction of approximately 6.0% in search runtime (from 6267.47 ms to 5893.55 ms), ensuring solution optimality while exploring fewer graph nodes. This validates the method's scalability for real-time applications.

Despite the promising results, the current model has limitations, specifically the lack of pavement roughness data (IRI) and the exclusion of real-time congestion or inertial energy losses from stop-and-go dynamics. Future work will focus on integrating real-time traffic APIs, classifying surface types to

adjust rolling resistance, and expanding the model to Electric Vehicles (EVs) to analyze regenerative braking potential on downhill segments.

In summary, this work reinforces that routing solutions based on static topography are powerful tools for developing cities, offering an accessible path toward more sustainable and energy-efficient intelligent transportation systems.

### ACKNOWLEDGMENTS

The authors would like to thank Prof. Michel Pires da Silva and the teaching assistant Matheus Emanuel da Silva for their assistance in this project.

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